

Healthcare Performance Analysis

Project Summary:

- ▶ This project focuses on transforming raw hospital data into a clear, meaningful, and decision-ready Power BI dashboard. The goal was to understand hospital performance, patient flow, departmental workload, treatment costs, patient feedback, and overall care quality.
- ▶ To achieve this, I built a 3-page interactive Power BI dashboard that gives hospital administrators a complete view of:
 - ▶ Hospital-level KPIs (staffing, occupancy, ER time, treatment costs)
 - ▶ Patient demographics and care insights
 - ▶ A detailed patient-level view for deep analysis
- ▶ With this dashboard, users can quickly identify bottlenecks, track performance across departments, and make better healthcare decisions backed by data.

Problem Statement & Objectives:

- ▶ Hospitals manage large amounts of patient and operational data. But without a unified reporting system, important questions often remain unanswered, such as:
 - ▶ Are we staffed enough for the number of patients we treat?
 - ▶ Why are ER wait times increasing?
 - ▶ Which age groups or departments need more attention?
 - ▶ How satisfied are patients with our services?
- ▶ The hospital lacked a single view to track patient flow, capacity, efficiency, cost, and patient experience — all in one place.

Objective:

- ▶ This project was designed to:
- ▶ Build a dashboard that gives a **complete 360° view of hospital performance.**
- ▶ Track key healthcare KPIs like **LOS, ER Time, Occupancy, ICU cases & Feedback.**
- ▶ Identify trends in **age groups, gender, departments, and states.**
- ▶ Highlight problem areas and improvement opportunities.
- ▶ Enable data-driven decision-making rather than assumptions

Dataset Overview:

- ▶ The dataset used for this project is the *Hospital Health Care Management Dataset*, which contains detailed records of patients, hospital operations, and healthcare service outcomes. It serves as the foundation for analyzing hospital efficiency, patient experience, resource utilization, and departmental performance.
- ▶ Total Records: 2,506
Total Columns: 20+ (across patient, clinical, operational & feedback data)
Missing Values: Minimal / Not significant (handled during data cleaning)

Key Statistical Observations:

- ▶ Average Treatment Cost: 161.41
- ▶ Average ER Time: 65.63 minutes
- ▶ Average LOS: 1.70 days
- ▶ Inpatient Count: 1,679
- ▶ Outpatient Count: 827
- ▶ Positive Patient Sentiment: ~95.6%
- ▶ These statistics give a strong starting point for analyzing patient flow, hospital workload, operational efficiency, and overall patient satisfaction.

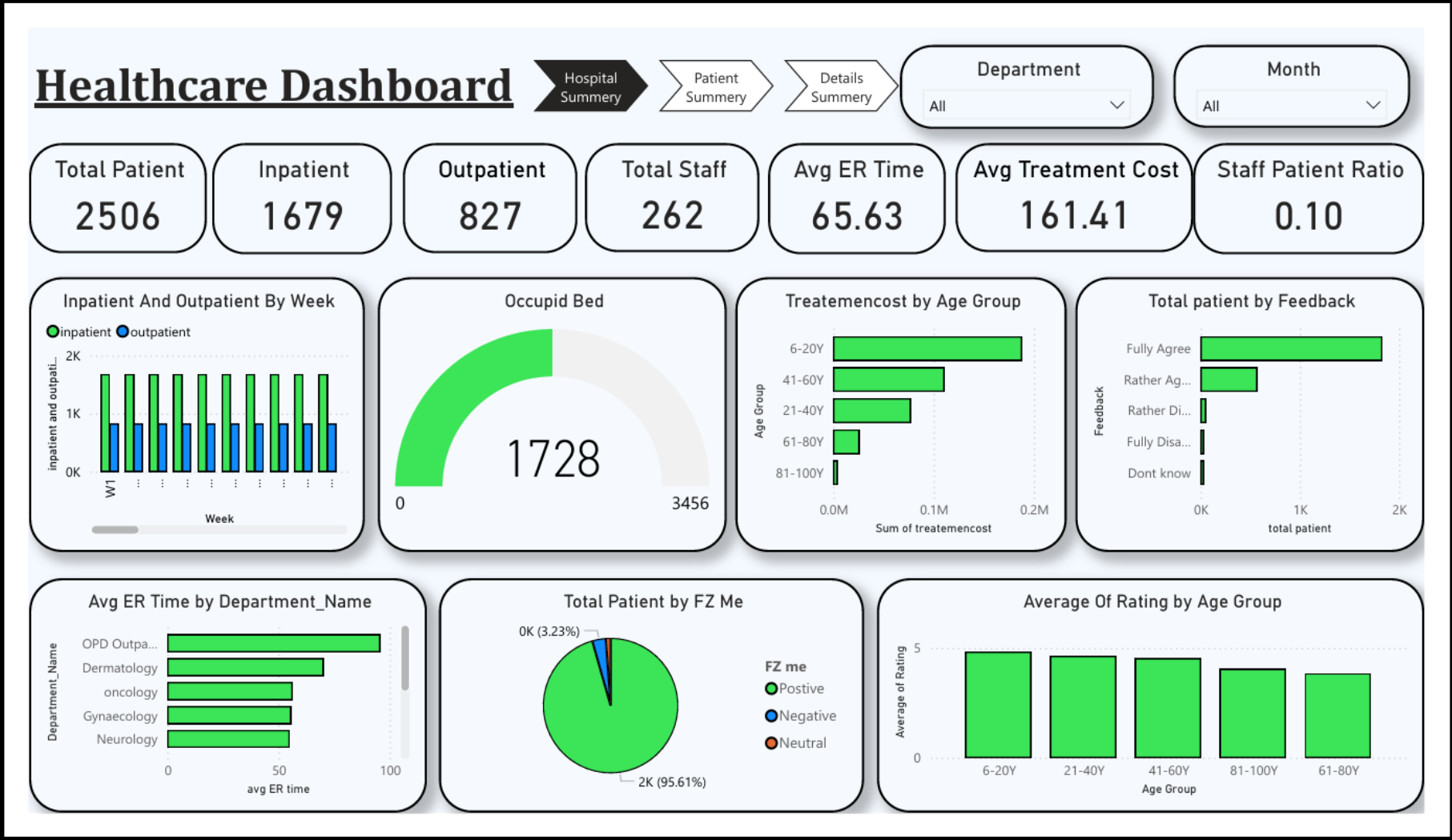
Tools & Approach:

To deliver this project, I followed a structured analytics workflow:

- ▶ Power Query - for cleaning, formatting, and preparing the data
- ▶ Power BI - for data modeling, creating DAX measures, and building the dashboard
- ▶ Interactive Filters & Drill-Throughs - to enable page navigation and patient-level analysis
- ▶ This approach ensured that the dashboard is not only visually appealing but also practical and easy for users to explore.

Key Insights From the Dashboard:

► Healthcare Dashboard:



Key Insights – Hospital Summary Dashboard

- ▶ The hospital served 2,506 patients, with inpatients (1,679) significantly higher than outpatients (827).
- ▶ Bed occupancy stands at 1,728 out of 3,456 (~50%), indicating sufficient capacity for additional patients.
- ▶ Average ER Time is 65.63 minutes, with OPD and Dermatology showing higher waiting times and requiring process improvement.
- ▶ The 6-20 age group contributes the highest treatment cost, followed by the 41-60 age group.
- ▶ Patient satisfaction is very high, with 95.6% positive feedback and only 3% negative responses.
- ▶ All age groups have strong average ratings (3.8-4.5), showing consistently good care quality across the hospital.



Patient Dashboard:

Patient Dashboard

Hospital
Summary

Patient
Summary

Details
Summary

Department

All

Month

All

Total Patient

2506

Admitted

870

Discharged

209

Avg Of LOS

1.70

ICU

82

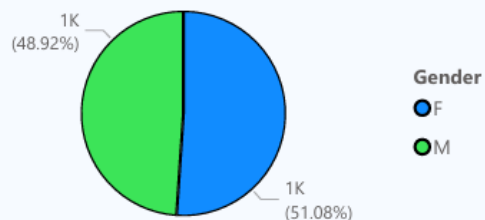
Death

86

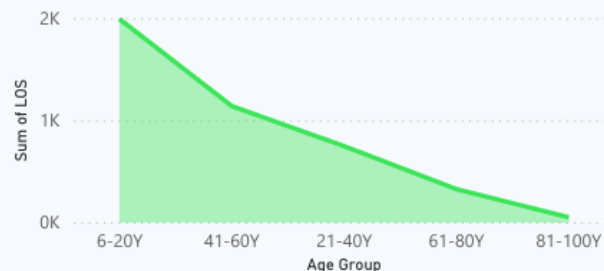
Occupied Bed

1728

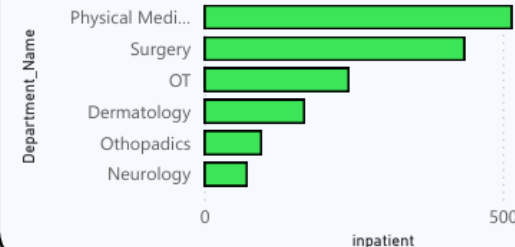
Total Patient by Gender



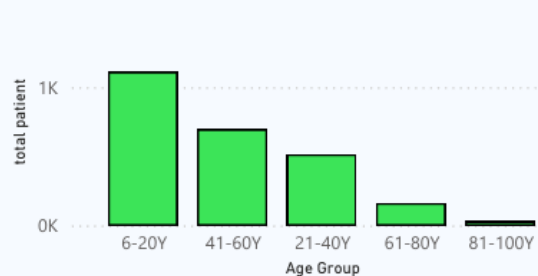
Sum of LOS by Age Group



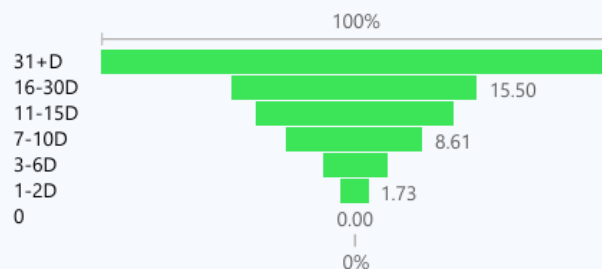
Inpatient by Department



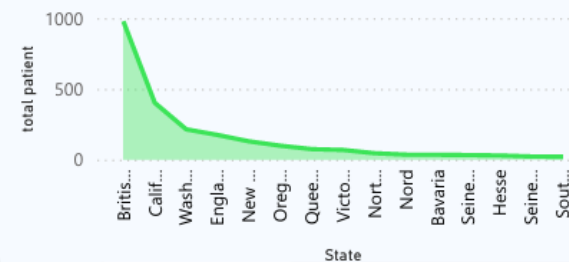
Total Patient by Age Group



Avg LOS by LOS Group



Total Patient by State



Key Insights – Patient Dashboard:

- ▶ A total of 2,506 patients were recorded, out of which 870 were admitted, 209 discharged, 82 treated in ICU, and 86 reported deaths.
- ▶ Gender distribution is almost equal with 51% male and 49% female patients, showing balanced patient demographics.
- ▶ The 6-20 age group has the highest number of patients, followed by 41-60 and 21-40 age groups.
- ▶ Length of Stay (LOS) increases with age, with the 6-20 group having the lowest LOS and 81-100 having the highest.
- ▶ Physical Medicine, Surgery, and OT account for the highest number of inpatient cases, indicating higher care demand in these departments.
- ▶ Majority of patients come from a few key states, with a steep drop for other regions, reflecting concentrated patient sourcing.

Details Dashboard:

Details Dashboard

Hospital
Summary

Patient
Summary

Details
Summary

Name	Gender	Sum of Age	City	State	Status	Department_Name	Sum of treatemencos
Chander, Nichole	F	70	Sunbury	British Columbia	Normal	OT	274.4
Chander, Nicolas	M	55	Sunbury	British Columbia	Normal	OT	152.6
Chander, Nina	F	46	Sunbury	British Columbia	Normal	OT	59.0
Chander, Omar	M	60	Sunbury	British Columbia	Normal	OT	208.5
Chander, Peter	M	64	Sunbury	England	Normal	OT	208.5
Chander, Priscilla	F	83	Sunbury	England	Normal	OT	5.6
Chander, Raul	M	53	Sunbury	Hesse	Normal	OT	16.3
Chander, Roger	M	51	Sunbury	South Australia	Normal	OT	5.6
Chander, Russell	M	74	Sunbury	Victoria	Normal	OT	2.5
Chander, Sharon	F	47	Sunbury	Hauts de Seine	Normal	OT	274.4
Chander, Shawn	M	82	Sunbury	South Australia	Normal	OT	4.1
Chander, Shawna	F	58	Sunbury	Hesse	Normal	OT	50.2
Chander, Sheena	F	53	Sunbury	England	Normal	OT	836.4
Chander, Tamara	F	56	Sunbury	California	Normal	OT	160.4
Chander, Tara	F	65	Sunbury	California	Normal	OT	31.3
Chander, Tasha	F	47	Sunbury	Berlin	Normal	OT	11.5
Chander, Terrence	M	47	Sunbury	Saxony	Normal	OT	4.1
Chander, Terry	M	76	Sunbury	Washington	Normal	OT	11.5
Chander, Tommy	M	43	Sunbury	New South Wales	Normal	OT	91.9
Chander, Tony	M	85	Sunbury	Washington	Normal	OT	4.3
Chander, Tonya	F	51	Sunbury	Queensland	Normal	OT	2.5
Chander, Warren	M	67	Sunbury	England	Normal	OT	43.6
Chander, Wayne	M	48	Sunbury	British Columbia	Normal	OT	6.3
Chander, Willie	M	76	Sunbury	Val de Marne	Normal	OT	73.5
Chander, Yolanda	F	50	Sunbury	Queensland	Normal	OT	4.0
Chandra, Abby	F	46	Sunbury	North Rhine-Westphalia	Normal	OT	274.4
Total		76426					4,04,488.1

Key Insights – Details Dashboard:

- ▶ The Details Dashboard provides patient-level data, enabling in-depth review of individual records including demographics, department, status, and treatment cost.
- ▶ It allows users to drill down into each patient's information to validate data, investigate anomalies, or review specific cases when required.
- ▶ Treatment costs vary significantly among patients, showing both low-cost and high-cost treatments within the same department, indicating diverse case complexity.
- ▶ Most patients listed in the sample belong to the OT (Operation Theater) department with a Normal status, suggesting a high volume of OT-based treatments.
- ▶ This view is particularly useful for audits, billing verification, exception handling, and patient-specific decision-making.

Recommendations to Improve Hospital Performance:

Based on the insights, here are a few improvement ideas:

- ▶ Reduce ER Waiting Time

- Introduce a “Fast-Track” system for low-severity patients

- Improve triage and appointment scheduling

- ▶ Optimize Department Resources

- Add more staff or time slots for busy departments like Surgery and Physical Medicine

- Use patient inflow patterns for smarter duty planning

- ▶ Strengthen Patient Experience

- Call back the small percentage of negative feedback cases

- Share monthly “You Asked - We Improved” updates to patients and staff

Conclusion:

- ▶ This healthcare dashboard makes hospital performance easy to monitor and understand. Instead of manually going through spreadsheets, hospital leaders can now:
 - Track KPIs in real-time
 - Identify areas that need attention
 - Improve patient experience
 - Make faster, data-driven decisions
- ▶ Overall, this project showcases how data analytics and Power BI can support better healthcare management, improve operational efficiency, and enhance the quality of patient care.